

ABE 721

Agricultural Systems Modelling and Simulation

Module 2: Models of Systems

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Contents

2 Models of Agricultural Systems **3**

2.1 2.1 Introduction and Terminologies 3

2.1.1 The Philosophy of Modelling 3

2.1.2 Core Systems Engineering Terminologies 4

2.1.3 Mathematical Representation of State-Space 5

2.2 2.2 Classification and Taxonomy of Scientific Models 7

2.2.1 1. Empirical vs. Mechanistic Models 7

2.2.2 2. Spatial Domain: Lumped vs. Distributed Models 9

2.3 2.3 The Engineering Modelling Process and Formulation 11

2.3.1 Deep-Dive Formulations (Agricultural Case Studies) 11

Chapter 2

Models of Agricultural Systems

The critical transition from merely recognizing the existence of an agricultural system (as fundamentally discussed in Module 1) to actively and mathematically simulating its behavior requires the rigorous formulation of a *model*. Agriculture represents one of the most mathematically complex intersections of biological, physical, chemical, and socio-economic systems on the planet.

This module delves deeply into the epistemological foundations of scientific modelling. We will define the strict terminologies employed in systems engineering, thoroughly explore the taxonomic classification of scientific models (from highly empirical black-box algorithms to robust mechanistic white-box models), and execute the rigorous mathematical calculus required for accurate model formulation.

2.1 2.1 Introduction and Terminologies

2.1.1 The Philosophy of Modelling

A mathematical model is, fundamentally, a deliberate abstraction of reality. It is never—and can never be—a perfect, one-to-one representation of the physical world. As the renowned statistician George E. P. Box famously stated in 1976: *"All models are wrong, but some are useful."*

In agricultural engineering, we model specifically to simplify the infinitely complex and chaotic realities of soil physics, plant physiology, and machine dynamics into mathematically tractable equations. These equations are designed to yield

predictive, explanatory, or normative (optimizing) insights.

Modelling explicitly requires the engineer to deliberately ignore certain aspects of reality. For instance, when modelling the required draft force of a tractor pulling a subsoiler, an engineer might ignore the aerodynamic drag force of the wind against the tractor chassis. While aerodynamic drag technically has a non-zero effect on the tractor's forward momentum, it is mathematically negligible compared to the massive forces of soil shear failure. The ultimate "art" and skill of engineering modelling lies in intelligently defining the *boundary* of abstraction.

2.1.2 Core Systems Engineering Terminologies

To communicate precisely and effectively in advanced systems engineering, a highly standardized vocabulary must be employed across all sub-disciplines (hydrology, machinery, crop science).

- **System:** A collection of interacting, interrelated, or interdependent physical elements forming a complex whole (e.g., a hydroponic greenhouse, a dairy cow's digestive tract, or a regional watershed basin).
- **System Boundary:** The conceptual, mathematical, and spatial line that strictly separates the system under investigation from its surrounding environment. Any mass, energy, or information crossing this boundary must be classified as an input or an output.
- **Environment:** Everything strictly outside the defined system boundary that can affect the system or be affected by it. The environment is the source of all exogenous variables.
- **Entity:** An individual, identifiable component or object within the system (e.g., a specific tractor, a water pump, or a single maize plant).
- **Attribute:** A quantifiable property or characteristic of an entity (e.g., the engine horsepower of a tractor, the volumetric capacity of a grain bin, or the Leaf Area Index (LAI) of a crop canopy).
- **State Variables (x):** A minimal, mathematically complete set of variables that

perfectly describes the exact condition or state of the system at any specific moment in time t. If the state variables are perfectly known at time t_0, and all future inputs are known, the exact future state of the system can be completely predicted (in a deterministic system). Examples include soil moisture content $\theta(t)$, instantaneous crop biomass $W(t)$, or reservoir volume $V(t)$.

- **Parameters (p or θ):** Constants that mathematically define the specific physical characteristics of the system equations. While state variables change dynamically over time during a simulation, parameters generally remain strictly constant. Examples include the saturated hydraulic conductivity of a specific soil type (K_s), the specific heat capacity of atmospheric air (C_p), or a machine's maximum theoretical field efficiency.
- **Exogenous Variables (Inputs, $\mathbf{u}(t)$):** Variables originating exclusively from the environment that drive, force, or influence the system. They are completely independent of the system's internal state. Examples include daily solar radiation (measured in MJ/m^2), precipitation, or operator-applied nitrogen fertilizer.
- **Endogenous Variables:** Variables mathematically generated entirely within the system boundary through the complex interaction of state variables, fixed parameters, and exogenous inputs.

2.1.3 Mathematical Representation of State-Space

In continuous-time dynamic engineering modelling, an agricultural system can generally be represented by a set of coupled ordinary differential equations (ODEs), universally referred to in control theory as the **State-Space Representation**.

For a highly complex, non-linear system, the generalized formulation is:

$$\frac{d\mathbf{x}(t)}{dt} = f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{p}, t) \tag{2.1}$$

Where:

- $\mathbf{x}(t)$ is the multi-dimensional state vector (e.g., $\mathbf{x} = [\text{Biomass}, \text{Soil Moisture}, \text{Nitrogen}]^T$).
- $\mathbf{u}(t)$ is the input forcing vector (e.g., $\mathbf{u} = [\text{Radiation}, \text{Rainfall}]^T$).

- **p** is the fixed parameter vector.
- t represents continuous time.
- f is a non-linear vector function defining the transient rates of change.

If the agricultural system can be approximated as *Linear and Time-Invariant (LTI)*, the state-space equations simplify beautifully into standard matrix algebra, allowing for rapid algorithmic solving:

$$\mathbf{x}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad \text{(State Equation)} \tag{2.2}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad \text{(Output Equation)} \tag{2.3}$$

Where **A** is the system matrix defining internal dynamics, **B** is the input matrix, **C** is the output matrix mapping states to observable variables $\mathbf{y}(t)$, and **D** is the direct feedthrough matrix (which is often zero in biological systems).

2.2 2.2 Classification and Taxonomy of Scientific Models

Before embarking on the complex mathematics of formulation, an engineer must classify the type of model required. Models in agricultural systems are generally constructed for one of four primary epistemological purposes: Explanatory (understanding *why*), Predictive (forecasting the *future*), Exploratory (asking *what if?*), or Normative (optimizing for the *best* outcome).

To achieve these purposes, scientific models are categorized across several distinct mathematical dimensions.

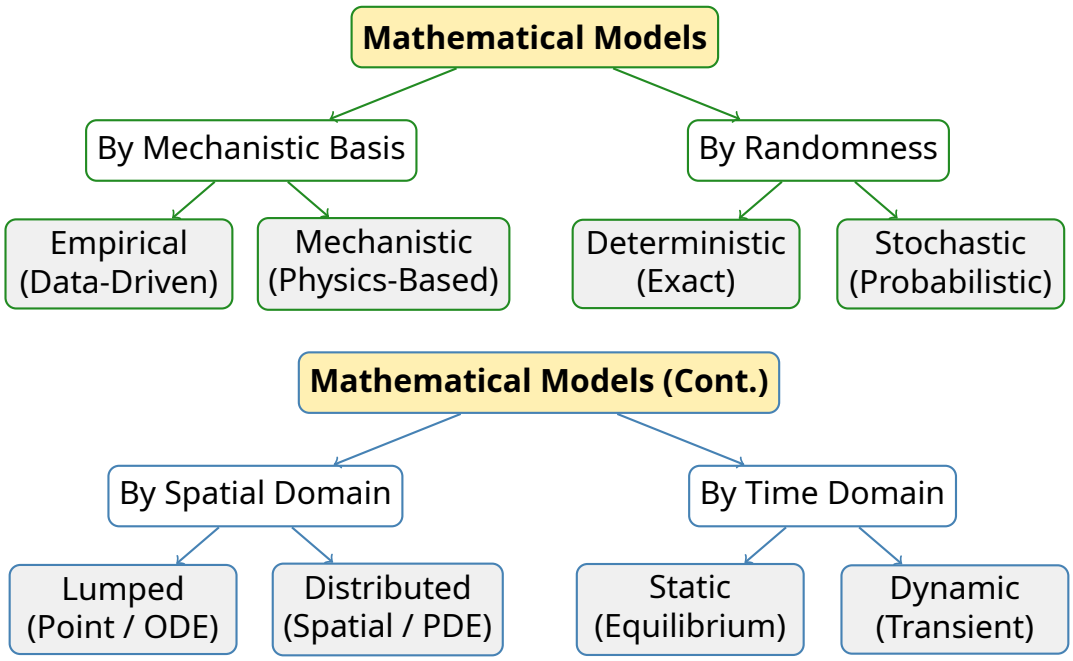


Figure 2.1: The comprehensive taxonomy of mathematical models utilized in Agricultural Systems Engineering.

2.2.1 1. Empirical vs. Mechanistic Models

Empirical Models (Black-Box Models)

Empirical models are derived strictly from observational data using statistical or machine-learning techniques. They mathematically describe the correlation between inputs and outputs without attempting to explain the underlying physical,

chemical, or biological mechanisms.

Traditional Example: Predicting crop yield (Y) based purely on total seasonal rainfall (R) and applied nitrogen (N) using a multivariate polynomial regression:

$$Y = \beta_0 + \beta_1 R + \beta_2 N + \beta_3 R^2 + \beta_4 (R \times N) + \epsilon \tag{2.4}$$

Advanced Example (Soft Computing): As detailed by Tayfur (2014) in the course syllabus, **Artificial Neural Networks (ANNs)** are highly advanced, non-linear empirical models. An ANN consists of input layers, hidden layers of ”neurons”, and an output layer. While an ANN can predict agricultural outputs with astonishing accuracy, it remains a ”black box” because the thousands of internal matrix weights (**W**) offer no comprehensible biological meaning.

Advantages: Rapid to develop if massive datasets are available; computationally very cheap to run. **Limitations:** Cannot be safely extrapolated outside the specific range of the training data. If a novel climate extreme occurs, or a new pest enters the region, the empirical model will fail catastrophically because it possesses no understanding of *why* plants grow.

Mechanistic Models (White-Box / Process-Based Models)

Mechanistic models are constructed upon the fundamental laws of nature: the conservation of mass, the conservation of energy, classical thermodynamics, and biological kinetics. They attempt to faithfully replicate the actual internal processes of the agricultural system.

Example: A mechanistic model of crop canopy growth does not simply correlate ”sunlight” to ”yield.” It models the exact physical attenuation of light through the leaf layers using the **Beer-Lambert Law**. The Photosynthetically Active Radiation (PAR) reaching a specific depth z in the canopy is given by:

$$I_z = I_0 \cdot \exp(-k \cdot LAI_z) \tag{2.5}$$

Where I_0 is the incident radiation at the top of the canopy, k is the specific light

extinction coefficient (a function of leaf angle distribution), and LAI_z is the cumulative Leaf Area Index down to depth z.

This absorbed energy ($I_0 - I_z$) is then biochemically converted into glucose (dry matter) using a specific radiation use efficiency parameter (ϵ):

$$\frac{dW}{dt} = \epsilon \times \left[I_0 \left(1 - e^{-k \cdot LAI}\right)\right] \tag{2.6}$$

Advantages: Extremely robust; capable of simulating unprecedented extreme conditions and novel climate change scenarios because the laws of physics do not change. **Limitations:** Extremely data-hungry; they require the calibration of dozens of difficult-to-measure physiological parameters (e.g., k, ϵ, maintenance respiration rates); computationally heavy.

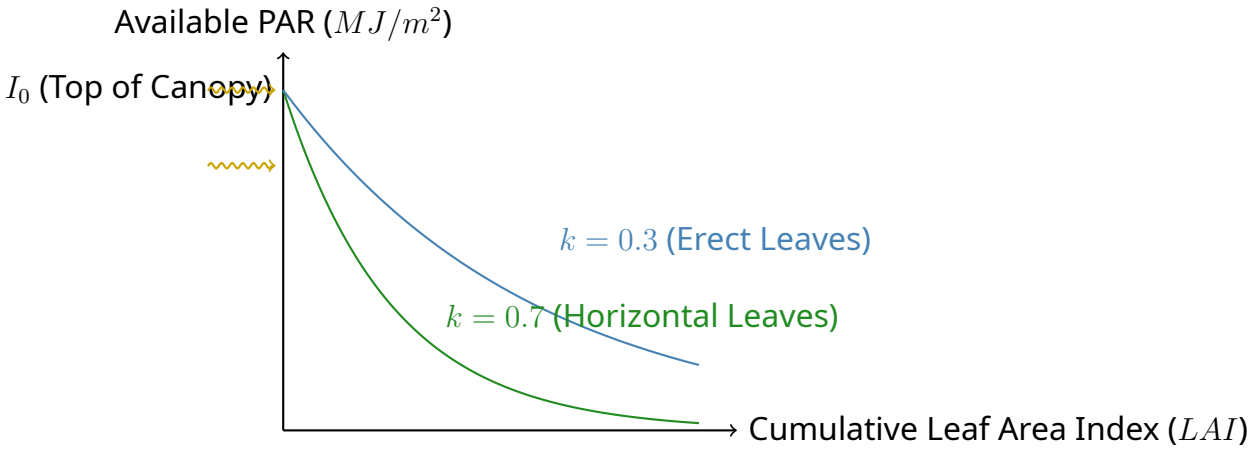


Figure 2.2: Mechanistic modelling of light interception using the Beer-Lambert Law. A crop with erect leaves (e.g., modern maize, $k = 0.3$) allows radiation to penetrate deeper into the canopy, resulting in higher overall photosynthetic efficiency compared to crops with horizontal leaves (e.g., beans, $k = 0.7$).

2.2.2 2. Spatial Domain: Lumped vs. Distributed Models

Lumped Models (Zero-Dimensional)

A lumped model entirely ignores spatial variation. The entire agricultural system (whether it is a 1-hectare field or a 1,000-hectare watershed) is mathematically compressed and treated as a single, homogenous, dimensionless point in space.

- **Mathematics:** The system is governed strictly by Ordinary Differential Equations (ODEs) where variables change only with respect to time ($\frac{dx}{dt}$).
- **Application:** Calculating the average daily soil moisture of a small, perfectly flat, uniform greenhouse bench.

Distributed Models (Multi-Dimensional)

A distributed model rigorously accounts for spatial heterogeneity. The physical farm or watershed is discretized (divided) into a massive 2D or 3D grid or mesh. State variables (like water pressure or nitrate concentration) are calculated distinctly for every single grid cell, allowing for interactions between neighboring cells.

- **Mathematics:** The system is governed by Partial Differential Equations (PDEs) where variables change with respect to both time and physical space ($\frac{\partial x}{\partial t}$ and $\frac{\partial x}{\partial z}$).
- **Application:** Precision agriculture; modelling how a localized fertilizer spill at coordinates (X_1, Y_1) leaches downwards and laterally through different soil horizons to contaminate groundwater at (X_2, Y_2) over a 30-day period.

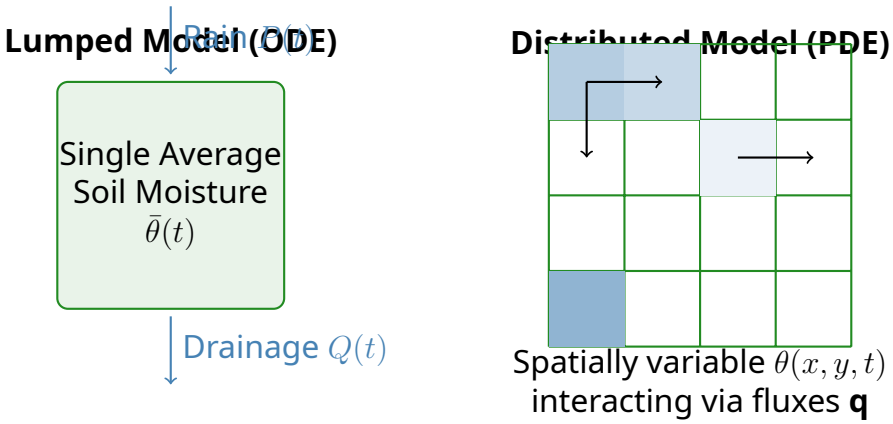


Figure 2.3: Visual comparison of spatial modelling paradigms. The Lumped model treats the field as a single uniform bucket. The Distributed model discretizes the field into a grid, allowing for precision mapping of moisture gradients and lateral subsurface flow.

2.3 2.3 The Engineering Modelling Process and Formulation

Developing a robust simulation model for an agricultural system is not merely an ad-hoc computer programming exercise; it is a highly rigorous, iterative scientific process. If the mathematical formulation is flawed at the conceptual stage, the final computer simulation will simply generate wrong answers at high speed.

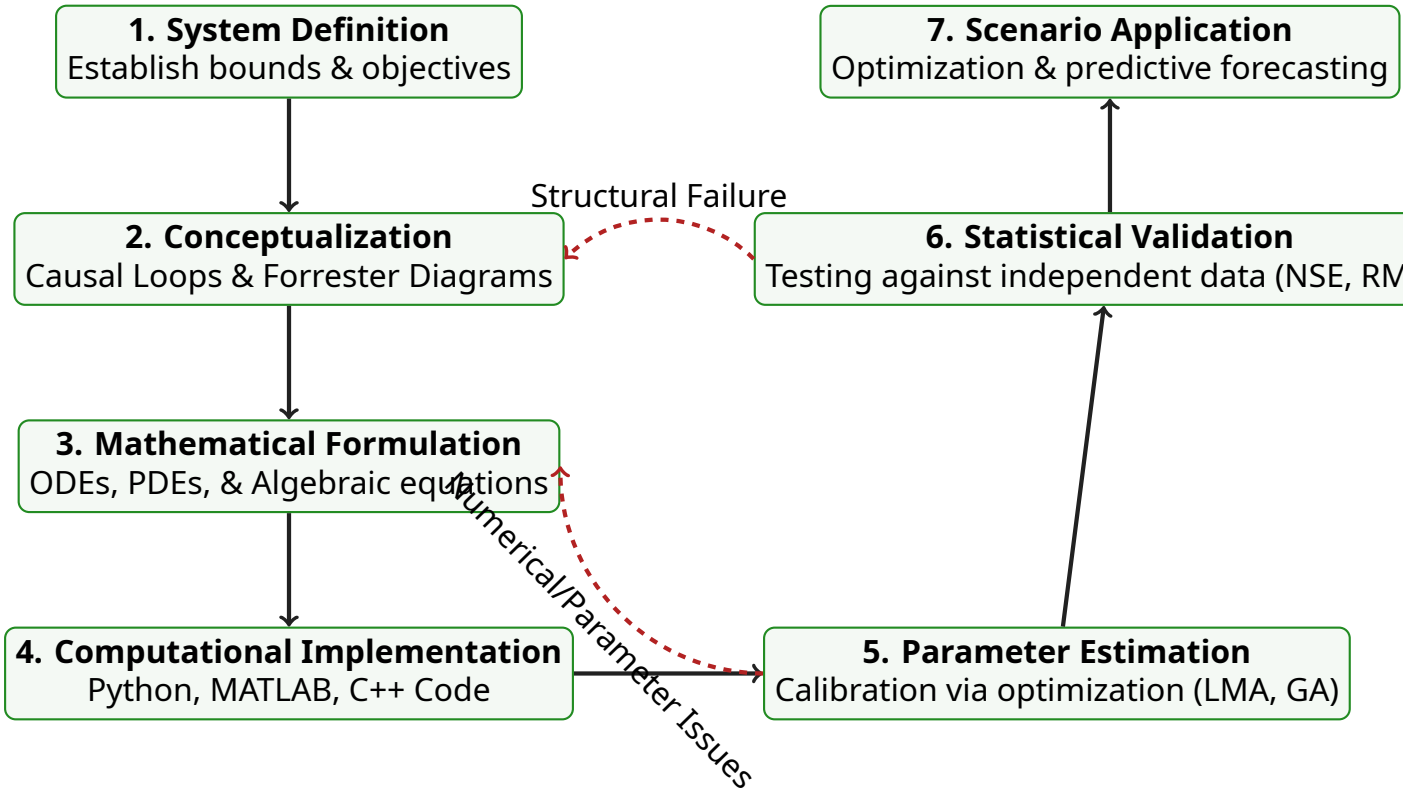


Figure 2.4: The Iterative Systems Modelling Process. Modelling is rarely linear. Failed statistical validation (Step 6) forces the engineer to return to Step 2 and re-evaluate the conceptual mechanics of the system.

2.3.1 Deep-Dive Formulations (Agricultural Case Studies)

This is the absolute core of agricultural systems engineering: translating chaotic biological, chemical, and mechanical physical realities into strict mathematics. We will explore three highly advanced case studies of mathematical formulation directly relevant to ABE 721.

Case Study A: Formulation of Soil Water Dynamics (The Richards Equation)

The Engineering Problem: We must rigorously model exactly how water moves vertically downwards through an unsaturated soil profile over time in order to design high-efficiency, precision drip-irrigation schedules.

Physical Principles and Derivation: 1. *The Principle of Conservation of Mass (Continuity Equation):* The transient change in volumetric water content (θ) within a specific soil control volume must exactly equal the net spatial flux (q) of water moving in and out, plus or minus any biological sink terms (S), which represents water sucked out by crop roots.

$$\frac{\partial \theta}{\partial t} = -\frac{\partial q}{\partial z} - S(z,t) \tag{2.7}$$

2. *The Darcy-Buckingham Law for Unsaturated Flow:* Water flux does not happen magically; it is driven strictly by potential energy gradients. Flux (q) is proportional to the gradient of the total hydraulic head (H). Crucially, in *unsaturated* agricultural soils, the hydraulic conductivity (K) is not a constant; it is highly non-linear and collapses exponentially as the soil dries out (as θ decreases).

$$q = -K(\theta)\frac{\partial H}{\partial z} \tag{2.8}$$

Because the total head $H = h(\theta) - z$ (where h is the matric suction head holding water to soil particles, and z is the gravitational head pointing strictly downwards), we substitute to get:

$$q = -K(\theta)\left(\frac{\partial h}{\partial z} - 1\right) \tag{2.9}$$

The Final PDE Formulation: Substituting the Darcy flux equation directly into the Continuity Equation yields the **1D Mixed-form Richards Equation**. This is the foundational, governing equation for all modern agricultural hydrology.

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left[K(\theta)\left(\frac{\partial h}{\partial z} - 1\right)\right] - S(z,t) \tag{2.10}$$

The van Genuchten Parameters: To actually solve this equation, we must mathematically define the complex relationship between K, h, and θ. The industry stan-

dard is the empirical van Genuchten (1980) equations:

$$\Theta = \frac{\theta - \theta_r}{\theta_s - \theta_r} = [1 + (\alpha|h|)^n]^{-m} \tag{2.11}$$

Where θ_r is residual moisture, θ_s is saturated moisture, and α, n, m are empirical parameters specific to the soil texture (e.g., clay vs. sand). These parameters must be estimated using the techniques discussed in Module 3.

Case Study B: Mechanical Formulation of Tractor Draft Force

Modelling is not restricted to biology and water; it is equally vital for machinery management (linking back to Module 1.3 Scheduling).

The Engineering Problem: To simulate the fuel consumption and field efficiency of a tractor pulling a tillage implement, we must mathematically model the required draft force.

Physical Principles: The ASABE (American Society of Agricultural and Biological Engineers) provides a robust, semi-empirical formulation for implement draft (D, measured in Newtons). The draft depends on soil texture, machine speed, and the depth/width of the tillage operation.

The ASABE standard formulation is a quadratic equation with respect to speed:

$$D = F_i \times \left[A + B(S) + C(S)^2\right] \times W \times T \tag{2.12}$$

Where:

- D: Implement draft requirement (N).
- F_i: Dimensionless soil texture adjustment parameter (e.g., 1.0 for fine clay, 0.7 for sandy loam).
- A, B, C: Machine-specific coefficients provided by ASABE tables (representing static friction, dynamic shear, and aerodynamic/inertial soil throw forces respectively).
- S: Operational speed of the tractor (km/h).

- W: Machine working width (m).
- T: Tillage depth (cm).

By coupling this draft force equation (D) with a thermodynamic model of the tractor's diesel engine efficiency, an engineer can dynamically simulate and optimize the exact fuel consumption (Liters/hectare) for a 500-hectare ploughing operation under varying soil conditions.

Case Study C: Thermodynamic Energy Balance of a Greenhouse

The Engineering Problem: We must computationally predict the transient internal air temperature (T_{in}) of a controlled environment agriculture (CEA) greenhouse to optimize the actuation of automated heating and ventilation systems.

Physical Principle: The First Law of Thermodynamics (Conservation of Energy). The instantaneous rate of change of thermal energy inside the defined greenhouse boundary equals the sum of all heat fluxes entering minus all heat fluxes leaving.

Let V be the volumetric airspace of the greenhouse, ρ_a be the density of air, and c_p be the specific heat capacity of air. The thermal mass of the internal air is therefore defined as $C_{air} = V \cdot \rho_a \cdot c_p$.

$$C_{air}\frac{dT_{in}}{dt} = Q_{solar} + Q_{heater} - Q_{conduction} - Q_{ventilation} - Q_{evapotranspiration} \tag{2.13}$$

Where the individual sub-models for the specific fluxes are defined as:

- **Solar Heat Gain (Q_{solar}):** $Q_{solar} = \tau \cdot A_g \cdot I_s$ (where τ is the specific transmittance of the plastic/glass cover, A_g is the floor area, I_s is incident solar irradiance).
- **Conduction Heat Loss ($Q_{conduction}$):** Governed rigorously by Fourier's Law of Heat Conduction. $Q_{conduction} = U \cdot A_c \cdot (T_{in} - T_{out})$ (where U is the overall heat transfer coefficient of the covering material, A_c is total cover area, and T_{out} is the exogenous weather temperature).
- **Sensible Ventilation Exchange (Q_{vent}):** $Q_{vent} = \dot{m}_{air} \cdot c_p \cdot (T_{in} - T_{out})$ (where

$\dot{m}_{air}$ is the controllable mass flow rate of air driven by the exhaust fans).

- **Latent Evaporative Cooling (Q_{ET}):** This represents the massive amount of thermal energy consumed and converted to latent heat as the crops actively transpire water. $Q_{ET} = \lambda \cdot E_T$ (where λ is the latent heat of vaporization of water, approx. 2.45 MJ/kg, and E_T is the canopy evapotranspiration rate).

Expanding the full formulation into a comprehensive ODE yields:

$$(V\rho_a c_p)\frac{dT_{in}}{dt} = \tau A_g I_s + Q_{heater} - UA_c(T_{in} - T_{out}) - \dot{m}_{air}c_p(T_{in} - T_{out}) - \lambda E_T \quad (2.14)$$

By formatting this non-linear equation into a standard state-space computational form, agricultural software engineers can design advanced PID (Proportional-Integral-Derivative) control algorithms to automatically actuate the Q_{heater} (furnace) and $\dot{m}_{air}$ (ventilation fans) to maintain T_{in} precisely at the optimal biological setpoint for maximum crop yield.

End of Module 2. Proceed to Module 3 for Qualitative and Quantitative Simulation Paradigms, and the Numerical Integration Techniques required to solve these ODEs and PDEs computationally.